

Statement of Purpose

A playable VGA-based video poker game that uses a random deck and keeps track of balance over multiple games

System Diagram

<https://excalidraw.com/#json=6SKW2Q2yreZfMF-uE6Hc5.2TP9WKid3IVFx3s1fr9X5g>

IO Pin Assignment Table

Tiny tapeout pin	Direction	Active level	Assignment
uo_out[0]	output	-	VGA Red bit 1
uo_out[1]	output	-	VGA Green bit 1
uo_out[2]	output	-	VGA Blue bit 1
uo_out[3]	output	-	VGA VSYNC
uo_out[4]	output	-	VGA Red bit 2
uo_out[5]	output	-	VGA Green bit 2
uo_out[6]	output	-	VGA Blue bit 2
uo_out[7]	output	-	VGA HSYNC
clk	input	Rising edge	25 MHz system and VGA pixel clock
rst_n	input	low	Asynchronous system reset
ena	input	high	Tiny Tapeout project enable
ui_in[0]	input	high	Hold card 1
ui_in[1]	input	high	Hold card 2
ui_in[2]	input	high	Hold card 3
ui_in[3]	input	high	Hold card 4
ui_in[4]	input	high	Hold card 5
ui_in[5]	input	high	deal/draw pushbutton

ui_in[6]	input	-	-
ui_in[7]	input	-	-
uio_in[7:0]	input	high	RNG seed
uio_out[7:0]	output	-	Driven zero
uio_oe[7:0]	output enable	low	Driven zero

Proposed Specification

The design simulates a [video poker](#) game machine, where the player can make wagers from an initial balance to play rounds of video poker and attempt to increase their balance. A VGA driven by the output pins will be used to display the game details, including the dealt and held cards, the player's balance, and the result of each round. The game supports an asynchronous reset that returns the game to its initial state. The player can use switches to select which cards to hold and debounced buttons to deal and draw cards. These switches and buttons are driven by the input pins.

The design simulates a five-card draw [video poker](#) game. The player begins with a balance of 100 credits, with each round requiring a fixed wager of 5 credits. Five cards are initially dealt, after which the player may select cards to hold using switches. The remaining cards are replaced when the player presses the draw button. Debounced push buttons are used to initiate the deal and draw actions.

After the draw, the resulting hand is evaluated and the player's balance is updated according to the following payout schedule for a 5-credit wager:

Hand	Payout
Royal Flush	4000
Straight Flush	250
Four of a Kind	125
Full House	45
Flush	30
Straight	20
Three of a Kind	15
Two Pair	10

Jacks or Better	5
All other hands	0

A VGA display driven by the output pins displays the dealt and held cards, current balance, and result of each round. The VGA controller operates using the 25 MHz input clock.

Cards are generated using a pseudorandom number generator (PRNG). The bidirectional I/O pins provide the PRNG seed, which is sampled when an asynchronous reset occurs. During development, switches connected to these pins allow the seed to be manually configured for testing and verification. In a production environment, these switches are not intended to be accessible to the player. The asynchronous reset also returns the game to its initial state, including restoring the player's balance to 100 credits.

The input pins are used for the player's switches and debounced push buttons, allowing the player to select cards to hold and initiate deal and draw actions.

Timeline for Completion

Sept. 24 — Proposal

- Finalize architecture, specifications, system diagram, FSM, and I/O assignments.

Oct. 4 — Core modules

- Implement VGA timing, LFSR, card registers, unique-card dealer, and initial tests.

Oct. 8 — Verilog design and test

- Integrate hold/draw logic, hand evaluator, game FSM, and basic VGA rendering.
- Complete directed Cocotb tests.

Oct. 12–22 — System integration

- Add input filtering, payouts, balance tracking, card graphics, and randomized tests.

Nov. 5 — Design evaluation

- Complete RTL verification, synthesis, utilization analysis, and timing checks.

Nov. 12–19 — Physical implementation

- Perform placement, routing, parasitic extraction, and gate-level verification.

Nov. 26 — System verification

- Resolve remaining issues and ensure all GitHub Actions pass.

Dec. 3 — Final submission

- Complete documentation and submit the Tiny Tapeout design.

Who does what

Nash — Game Logic

- Game state machine & round flow
- Game orchestration / control logic
- Card dealing & deck management
- Poker hand evaluation & payouts

Luke — I/O & Hardware Interface

- Player input handling
- Button debouncing
- Pseudorandom number generator & seeding
- VGA controller & game display